

# Technical Note — MPI Risk Engine v03a (Expanded, Audit-Ready)

## 1) Background & Purpose

The MPI Risk Engine v03a produces blended project success probabilities by combining a Bayes prior leg and a Cox survival leg. This patched version introduces audit-friendly improvements, including province–sector shrinkage, expanded coefficients, and clearer control knobs. It is designed for reproducibility and transparent auditing.

## 2) Inputs

- Coefficient files: bayes\_lr\_regenerated\_coefficients\_expanded.csv, cox\_refit\_coefficients\_timesplit\_expanded.csv
- MPI input file: mpi\_2024\_input.xlsx (must include project metadata and costs)

## 3) Configuration Knobs

- PROVSEC\_ENABLED (default true): enable province–sector interaction
- PROVSEC\_ALPHA (default 0.6): shrinkage factor; 1.0 = no shrink
- BLEND\_BAYES\_W (default 0.5): weight on Bayes leg in blended probability
- S0\_T (default 0.545332): Cox survival baseline at 5 years

## 4) Methodology

### **Bayes Leg:**

Builds features from cleantech, cost quintile, group, province, sector, start year, FOAK, and Greenfield. Applies province–sector shrinkage when enabled. Computes  $p_{\text{bayes}}$  via Naive Bayes log-odds.

### **Cox Leg:**

Computes linear predictor  $\eta$  using expanded coefficients (province, sector, cleantech, cost\_percentile, FOAK, Greenfield). Converts to  $\text{risk\_score} = \exp(\eta)$ . Survival probability scaled to 5-year baseline.

### **Blending:**

$\text{blended\_prob} = w * p_{\text{bayes}} + (1 - w) * p_{\text{cox}}$ , with  $w = \text{BLEND\_BAYES\_W}$ .

### **Priority Index & Ranking:**

$\text{priority\_index} = \text{blended\_prob} / \text{years\_remaining}$ ; urgency scale normalized 0–1;  $\text{power\_ranking} = 0.60 * \text{blended\_prob} + 0.40 * \text{urgency\_scale}$ .

## 5) Outputs

- mpi\_2024\_scored.csv and mpi\_2024\_scored.xlsx
- $p_{\text{bayes}}$ ,  $p_{\text{cox}}$ , blended\_prob

- risk\_score, priority\_index, urgency\_scale\_(0-1), power\_ranking

## **6) Replication Procedure**

- Provide coefficient CSVs and input file mpi\_2024\_input.xlsx
- Set environment variables if customizing knobs
- Run mpi\_risk\_engine\_v03\_expanded.py
- Archive input and output files
- Record configuration meta (console prints PROVSEC\_ALPHA, BLEND\_BAYES\_W, coefficient paths)

## **7) QA / Audit Checks**

- Verify Bayes shrinkage applied to province–sector interactions
- Check that probabilities remain in [0,1]
- Confirm EV scaling and blended\_prob match inputs
- Validate console META printout with archived outputs

## **8) Limitations**

The engine does not retrain models; it applies fixed coefficients. Shrinkage and blend weights are stylized, intended to prevent swamping and allow sensitivity tests. Results highlight distributional patterns and rankings, not forecasts.